

Dynamic Modeling of Bellows-Actuated Continuum Robots Using the Euler–Lagrange Formalism

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Abstract—In the previous decade, multiple useful approaches for kinematic models of continuum manipulators were successfully developed. However, dynamic modeling approaches needed for fast simulations and the development of model-based controller design are not powerful enough yet—especially for spatial manipulators with multiple sections. Therefore, a practicable lumped mass model similar to common dynamic models of rigid-link manipulators is needed, which can be used for simulations and model-based control design. The model incorporates mechanical interconnections of parallel and serially connected bellows and uses constant curvature kinematics and its analytical derivatives to balance forces and energies in a global reference frame. The parameters of the resulting model are identified with measurements before the simulation results are experimentally validated. The obtained dynamic model can be used to both simulate the manipulator dynamics and calculate the inverse dynamics needed for model-based controller design or path planning.

Index Terms—Continuum kinematics, dynamic model, inverse dynamics, lumped model, redundant robots.

I. INTRODUCTION

CONTINUUM robots and hyperredundant robots inspired by snakes, octopus arms, eels, or trunks have gathered increasing attention in the past decade of robotics research. With their continuous backbones and their inherent flexibility, new fields of application are being developed, where rigid-link robots with joints are not beneficial, i.e., in environments with multiple barriers and edges. On the other hand, industrial rigid-link robotics have been developed for a long period of time, and its methods for modeling, control, and path planning enable multiple applications [1].

Useful kinematic models for continuum robots were developed almost ten years ago, especially the constant curvature approach [2] has been commonly used [3]. In order to investigate and derive kinematics of continuum robots, also static mechanical models in both planar and spatial manner were developed, using beam mechanics [4], [5], Cosserat rod [6], [7], elliptic integrals [8], modal approaches [9], or even a lumped model [10].

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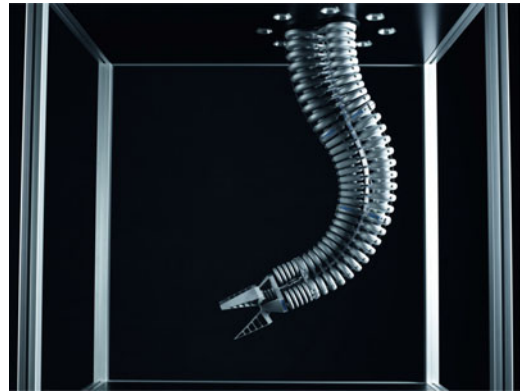


Fig. 1. Festo's Bionic Handling Assistant is a continuum manipulator with three sections and a gripper. This continuum manipulator changes its pose by deflating or inflating its nine parallel and serial arranged bellows. (Source: Festo AG).

However, due to the complexity, all these mechanical models were considered in static equilibrium only.

Dynamic models for continuum manipulators or the related hyperredundant manipulators were investigated already in the early stages to find a closed-form description [11], but compared with kinematic models, the progress of dynamic models is still lacking [9].

Only a few are using Cosserat rod models for dynamic simulations of single-section manipulators [12] and manipulators with multiple sections [13], but the resulting partial differential equations are not suitable for fast online simulations or controller design but rather for general simulations of soft robotics under friction and external loads.

A close approximation of geometrically exact solutions of hyperflexible and hyperredundant manipulators can be achieved by discretizing continuous models resulting multiple discs or slices. Using mass and stiffness matrices of the slices, the dynamic equations of the manipulator are obtained by integrating over multiple discs. Early works cover the spatial dynamic modeling of single-section manipulators [14] and the planar modeling of multisectional manipulators [15], [16]. For manipulators with many modules, also recursive algorithms for spatial modeling [17] were developed. More recent works obtain the dynamic models of slice approximated manipulators using modal transformations and Lagrangian [18] or the principle of virtual work [19]. Although these models might be well suited for simulations, they are not applicable to model-based control design.

Lumped models are common for rigid-link robots as they provide rather simple but powerful models suitable for fast simulations and model-based control. However, lumped mass approaches for continuum robots are still hard to find. On the

TABLE I
NOMENCLATURE

Symbol	Variable name
n	number of sections
q_{ik}	actuator state of section i and bellows k
$\mathbf{q}_i$	actuator states of section i
q_j	actuator state j
$\mathbf{k}_i$	configuration state of section i
ϕ_i	orientation of section i (config. state)
κ_i	curvature of section i (config. state)
ℓ_i	elongation of section i (config. state)
ψ_i	bending angle of section i
${}^j\mathbf{H}_i$	homogen. transf. matrix of frame i in ref. frame j
${}^j\mathbf{R}_i$	rotation matrix of frame i in reference frame j
$\mathbf{r}$	position ($\mathbf{r}$ in homogeneous representation)
$\mathbf{g}$	gravity vector
$\mathbf{F}$	force ($\mathbf{F}$ in homogeneous representation)
$M_{i,\text{bend}}$	bending momentum of section i
m_i	mass of section i
$\mathbf{J}_{Si}$	moment of inertia of mass m_i
$\boldsymbol{\omega}_i$	rotational velocity of mass m_i
T	kinetic energy
U	potential
Q	generalized force
$\mathbf{M}$	mass matrix of dynamic equation
$\mathbf{C}$	Coriolis matrix of dynamic equation
$\mathbf{N}$	force vector of dynamic equation
$\boldsymbol{\tau}$	actuation force vector of dynamic equation
Index	Subscript and superscript description
w	world
ih	head of section i
ib	base of section i
Si	center of mass of section i
Bik	center of bellows area of section i and bellows k
e	active (effective) force
a	actuation force
p	passive force
*	general placeholder for similar items

one hand, this might be caused by the idea that only continuum models with distributed parameters instead of concentrated parameters resulting in partial differential equations can describe the dynamic complexity of continuum manipulators. On the other hand, most continuum manipulators are not operated close to dynamic boundaries and thus require only kinematic and static mechanical models. However, a few lumped manipulator models for continuum manipulators were developed so far, most of them in the last years.

Lumped approximations for hyperredundant octopus-like manipulators were developed, e.g., to describe the planar motion of an arm by using decoupled actuator models with a diagonal mass matrix resulting in a decoupled differential equation system obtained by Lagrange formalism [20]. Another algorithm that describes the spatial net motion of a platform with multiple arms was derived by recursively computing reaction forces of actuator segments of multiple arms taking into account the dynamics of the adjacent actuators [21]–[23]. Although forces are exchanged, the dynamic actuator models itself are still decoupled. A planar model for a tendon-driven manipulator was obtained by defining a high number of point masses. The large but sparse stiffness and mass matrices combined with a Rayleigh damping approach could be used for a modal decomposition and modal control [24].

Even less approaches for bellows or muscle-driven continuum manipulators exist. For a single section, a planar model with three masses along the backbone and spring–damper–actuator systems along the actuators was presented using the Euler–Lagrange formalism [25]. However, no dynamic results were provided, and the static simulations were not convincing.

However, the advantages of lumped manipulator models still remain: They are computationally cheap and the methods already developed for flexible multibody systems [26] and rigid-link manipulators [1] can be easily transferred and applied to other lumped models, like inverse dynamics control. Furthermore, the control strategy of bellows actuated manipulators that are limited by slow dynamics on the one hand and undesired interaction of the actuators during faster motions on the other hand could be improved using inverse dynamics control.

Hence, in this paper, an explicit lumped dynamic manipulator model is presented that provides a useful analytical approximation and considers dynamic couplings of parallel and serial actuators using the Euler–Lagrange formalism. The presented model is applicable to continuum manipulators with intrinsic actuators that can be described with kinematics based on homogeneous transformations like constant curvature kinematics [3], variable curvature [27], or piecewise constant curvature [28] kinematics. Compared with a previous published conference paper [29], also rotational energies are considered now and an outline toward identification of parameters is provided as well.

In order to derive the model, for each section of a multi-sectional continuum manipulator, lumped point masses were chosen. Based on a mass–spring–damper system of a single bellows [30], a lumped section model with spring–damper representation for each actuator is developed. Using the kinematic model, local forces are transformed into the world reference frame, such that an Euler–Lagrange formalism can be used [1]. Then, the theoretical model is applied to Festo’s Bionic Handling Assistant [31], a bellows-actuated continuum manipulator with three sections (see Fig. 1).

After introducing the kinematic model of the multisectional constant curvature manipulator in Section II, the dynamic model is derived in Section III: First, the general model is presented; then, the generalized forces, the potential, and the kinetic energy are derived. Furthermore, the analytical derivatives of the homogeneous transformations of the kinematic model need to be computed before the resulting dynamic model with its properties can be presented. In Section IV, the dynamic parameters are identified on the example manipulator before the model is reduced in Section V. After an experimental validation of the model in Section VI, a final conclusion is drawn in Section VII.

II. KINEMATIC MODEL

A bellows-driven continuum manipulator in general has three actuators per section (see Fig. 2c). By serially connecting the sections, a manipulator with multiple sections can be constructed, i.e., with three sections and nine bellows in total (see Fig. 2a). In order to describe the relative and absolute positions of the end effector and the actuators, a kinematic model is needed. For this purpose, different coordinate frames are introduced: actuator space, configuration space, and task space.

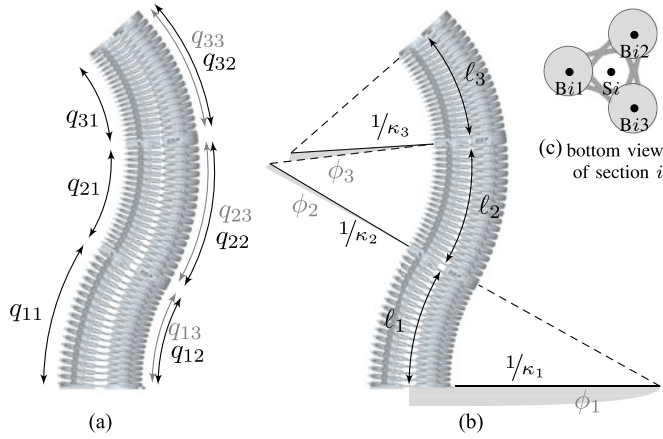


Fig. 2. (a) Actuator states $\mathbf{q}$ can be measured and controlled, while (b) configuration states $\mathbf{k}$ describe the shape of the manipulator.

First, the different spaces are introduced before the transformation of the forward kinematics is explained.

A. Kinematic Coordinate Frames

In actuator space, the actuator states

$$\mathbf{q} = [q_{11} \ q_{12} \ \dots \ q_{n3}]^T \quad (1a)$$

$$= [q_1 \ q_2 \ \dots \ q_{\{3n\}}]^T \quad (1b)$$

of n sections representing the length l of each bellows can be directly measured and controlled (see Fig. 2a). Two indexing methods allow a bellows- and section-wise notation (1a) of the actuator state q_{ik} with section index $i \in \{1, \dots, n\}$ and bellows index $k \in \{1, 2, 3\}$ and a global actuator-wise notation (1b) of all vector elements q_j with global actuator index $j \in \{1, \dots, 3n\}$.

In configuration space, the configuration states $\mathbf{k}$ characterize an actuator-independent model of the manipulator by describing its general shape. These configuration states are provided by the kinematic model. A well-suited kinematic model for continuum manipulators with multiple sections is the constant curvature approach [2], [3] that defines the configuration states (see Fig. 2b) as the section's backbone length ℓ , its curvature κ , and the orientation ϕ of the deformation for each section i by

$$\mathbf{k}_i = [\phi_i \ \kappa_i \ \ell_i]^T. \quad (2)$$

For highly conical sections, the model can also be extended to a piecewise constant curvature approach [28]. The curvature κ_i is reciprocal to the section's bending radius, while the product of backbone length and curvature define the bending angle ψ_i

$$\psi_i = \kappa_i \ell_i. \quad (3)$$

Industrial manipulators are programmed in the end effector's task frame. These task frames often use Cartesian coordinates as they are very intuitive for humans. Therefore, Cartesian coordinate frames $\mathcal{R}$ for the end effector $\mathcal{R}_e$, for the head $\mathcal{R}_{ih}$ and base $\mathcal{R}_{ib}$ of the manipulator's serially linked sections, and a world reference frame $\mathcal{R}_w$ are introduced (see Fig. 3). As the manipulator cannot twist itself, the task space coordinates

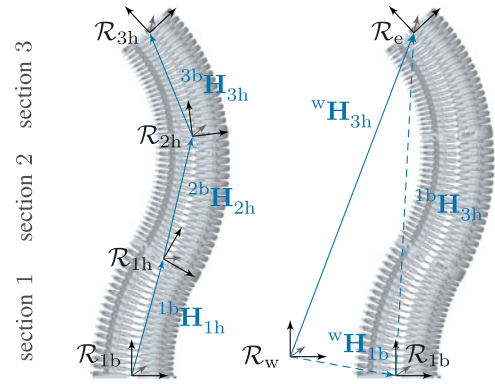


Fig. 3. Serial combination of local task space reference frame defined at each section's head result in a Cartesian task space representation of a continuum manipulator's end effector in a world reference frame.

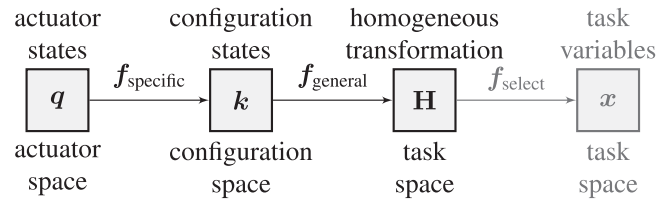


Fig. 4. Kinematic spaces and their corresponding transformations.

consist of three position coordinates and two orientation angles

$$\mathbf{x} = [x \ y \ z \ \varphi_x \ \varphi_y]^T \quad (4)$$

which can also be expressed by a homogeneous transformation matrix $\mathbf{H}$ including a rotation matrix $\mathbf{R}$ and a translation vector $\mathbf{o}$

$${}^j\mathbf{H}_i = \begin{bmatrix} {}^j\mathbf{R}_i & {}^j\mathbf{o}_i \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4} \quad (5)$$

that describes the relative pose between two Cartesian coordinate frames i and j and allows a derivation of any task coordinate combination, e.g., position only or complete pose.

B. Kinematic Transformations

After defining the coordinate frames, the mappings between these frames need to be determined (see Fig. 4): a specific mapping $\mathbf{f}_{\text{specific}}$ that calculates configuration states $\mathbf{k}$ in response to the according actuator states $\mathbf{q}$ and a general mapping $\mathbf{f}_{\text{general}}$ that transforms configuration states into task states.

The specific mapping $\mathbf{f}_{\text{specific}}$ describes the relation between actuator states and the configuration states for each section i . Using the constant curvature approach [3], the configuration states (2) are obtained from actuator states (1) by

$$\mathbf{f}_{\text{specific}} : \mathbf{k}_i = \mathbf{k}_i(q_{i1}, q_{i2}, q_{i3}) = \mathbf{k}_i(\mathbf{q}_i). \quad (6)$$

This manipulator specific mapping may differ for each manipulator. For instance, applying the constant curvature approach to a section of the Bionic Handling Assistant with median distance r_{Bi} from section back bone to bellows back bone results

in the transformation equations¹ [28]

$$\phi_i = \arctan2\left(\sqrt{3}(q_{i3} - q_{i2}), q_{i2} + q_{i3} - 2q_{i1}\right), \quad (7a)$$

$$\kappa_i = 2 \frac{\sqrt{q_{i1}^2 + q_{i2}^2 + q_{i3}^2 - q_{i1}q_{i2} - q_{i2}q_{i3} - q_{i3}q_{i1}}}{(q_{i1} + q_{i2} + q_{i3}) r_{Bi}}, \quad (7b)$$

$$\ell_i = \frac{1}{3}(q_{i1} + q_{i2} + q_{i3}). \quad (7c)$$

If all three actuators have the same length such that the section is not bent but straight, the curvature κ_i is zero, and the orientation ϕ_i is not defined.

The general mapping

$$\mathbf{f}_{\text{general,loc}} : {}^{ib}\mathbf{H}_{ih} = {}^{ib}\mathbf{H}_{ih}(\mathbf{k}_i) \quad (8)$$

between the base (index b) and the head element (index h) of a single section i is based on the homogeneous transformation (5) and defines the pose of each section's head. As the configuration states are manipulator-independent, the definition of the general transformation is valid for all continuum manipulators with constant curvature and is obtained by

$${}^{ib}\mathbf{H}_{ih} = \begin{bmatrix} c_\phi^2(c_{\kappa\ell} - 1) + 1 & s_\phi c_\phi(c_{\kappa\ell} - 1) & c_\phi s_{\kappa\ell} & \frac{c_\phi(1 - c_{\kappa\ell})}{\kappa} \\ s_\phi c_\phi(c_{\kappa\ell} - 1) & c_\phi^2(1 - c_{\kappa\ell}) + c_{\kappa\ell} s_\phi s_{\kappa\ell} & \frac{s_\phi(1 - c_{\kappa\ell})}{\kappa} & \\ -c_\phi s_{\kappa\ell} & -s_\phi s_{\kappa\ell} & c_{\kappa\ell} & \frac{s_{\kappa\ell}}{\kappa} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

using the abbreviations

$$c_\phi = \cos \phi_i, \quad s_\phi = \sin \phi_i, \quad (10a)$$

$$c_{\kappa\ell} = \cos \kappa_i \ell_i, \quad s_{\kappa\ell} = \sin \kappa_i \ell_i. \quad (10b)$$

If the curvature κ_i is zero and the section's orientation ϕ_i is not defined, (9) becomes

$${}^{ib}\mathbf{H}_{ih} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \ell_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (11)$$

implying an upright section that is only characterized by its length ℓ_i .

The general mapping $\mathbf{f}_{\text{general}}$ of the tool at the head of the upper section $i = n$ in relation to a world reference frame with

$$\mathbf{f}_{\text{general,glob}} : {}^w\mathbf{H}_{ih} = {}^w\mathbf{H}_{ih}(\mathbf{k}) \quad (12)$$

is obtained by sequentially stacking the serial sections on top of each other. In this case, the section's base coordinates are identical with the head coordinates of the section below such that

$$\{{}^{i-1}\}^h\mathbf{H}_{ib} = \mathbf{I}_4 \quad (13)$$

with $\mathbf{I}_n$ being the identity matrix of dimension n . Therefore, the transformation of a vector in head coordinates of a section i

into the reference coordinate system $\mathcal{R}_w$ (12) is obtained by calculating the product

$${}^w\mathbf{H}_{ih}(\mathbf{k}(\mathbf{q})) = \underbrace{{}^w\mathbf{H}_{1b}}_{\text{const.}} \prod_{\xi=1}^i {}^{\xi h}\mathbf{H}_{\xi b}(\mathbf{k}_\xi(\mathbf{q}_\xi)) \quad (14)$$

of all local sectional transformations (9) and the constant initial transformation ${}^w\mathbf{H}_{1b}$ (see Fig. 3).

III. DYNAMIC MODEL

In order to derive a dynamic model of a continuum manipulator with multiple sections that can be used for model-based control and that includes all main characteristics like couplings between the bellows and is still computationally efficient, the distributed mass needs to be discretized. The dynamic model for discrete masses is derived using the Euler–Lagrange equations

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} = Q_j \quad (15)$$

for the general coordinates q_j with the kinetic energy T , the velocity-independent potential U , and the corresponding generalized forces Q_j . Having n discrete masses m_i , their total kinetic energy

$$T = \frac{1}{2} \sum_{i=1}^n m_i {}^w\mathbf{r}_{Si}^T {}^w\dot{\mathbf{r}}_{Si} + \frac{1}{2} \sum_{i=1}^n {}^w\boldsymbol{\omega}_i^T {}^w\mathbf{J}_{Si} {}^w\boldsymbol{\omega}_i \quad (16)$$

is the sum of their translational energies depending on the linear velocity of each mass ${}^w\dot{\mathbf{r}}_{Si}$ and the rotational kinetic energies depending on the rotation rate $\boldsymbol{\omega}_i$. The potential U includes gravitational potentials and rotational spring potentials due to bending. Without loss of generality, translational spring forces due to extension are covered in the generalized forces

$$Q_j = \sum_{\mu=1}^{n_f} {}^w\mathbf{F}_{e,\mu} \frac{\partial {}^w\mathbf{r}_\mu}{\partial q_j} \quad (17)$$

that consist of the projection of n_f active forces ${}^w\mathbf{F}_e$ at the position ${}^w\mathbf{r}$ excluding potential forces considered in U .

A. General Model Derivation

The discrete masses representing the distributed mass and the moments of inertia have to be chosen. Therefore, it is assumed that

- each section i has one concentrated mass m_i located at a locally fixed position $\mathbf{r}_{Si}$ in the section's head reference frame (see Fig. 5), and
- the moments of inertia ${}^{ih}\mathbf{J}_{Si}$ of the discrete masses remain constant in the local coordinate frame and are approximated as a hollow cylinder.²

Based on these assumptions, the general idea of the model derivation is presented next.

As the manipulator consists of multiple sections, a modular section model is derived first that can be serially coupled later.

¹ $\arctan2$ is defined s.t. $\phi = \arctan2(\sin \phi, \cos \phi)$ with $\phi \in (-\pi, \pi]$.

²This assumption will be discussed in the identification section (see Section IV-B).

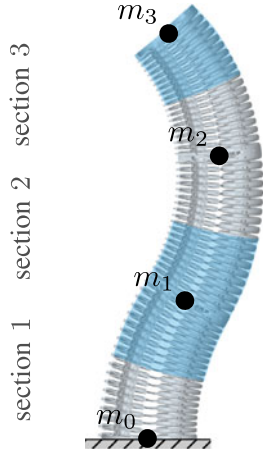


Fig. 5. Approximation of distributed masses as point masses located at each section's head. The top most point mass also includes possible gripper and load masses; m_0 is fixed.

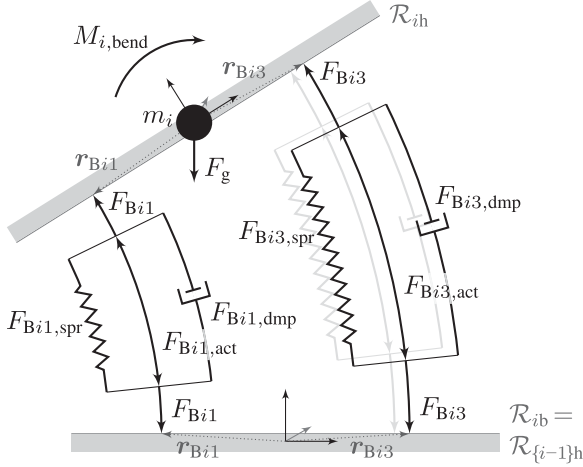


Fig. 6. Section model with one concentrated mass and three bellows represented by a spring–damper–actuator system (note: 2nd actuator in background).

Therefore, a push–pull actuator–spring–damper model of a single bellows [30] that characterizes only the longitudinal motion is extended to a section model containing three parallel bellows: each section consists of three mass-free bellows modeled as push–pull actuator–spring–damper systems and a combined point mass at the section's head (see Fig. 6). The gravity force F_g applies at the center of mass of the i^{th} section r_{Si} that has a fixed relative position within the section's head

$${}^{ih}r_{Si} = \text{const.} \quad (18a)$$

$${}^{ih}r_{Bi k} = \text{const.} \quad (18b)$$

as well as the relative positions of the bellow's contact points $r_{Bi k}$. Using the homogeneous transformation (14), position vectors and forces³ are transformed

$${}^w r_{i*} = {}^w H_{ih} {}^{ih} r_{i*} \quad (19)$$

from the local head coordinate system $\mathcal{R}_{ih}$ of the section i to the global reference system $\mathcal{R}_w$. The partial derivatives and

³If position vectors are multiplied with a homogeneous transformation, they are extended to $[\mathbf{r}^T \ 1]^T$, while force vectors are extended to $[\mathbf{F}^T \ 0]^T$.

time derivatives of position vectors that are fixed in the head coordinate system (18) are derived to

$$\frac{\partial {}^w r_{i*}}{\partial q_{ik}} = \frac{\partial ({}^w H_{ih} {}^{ih} r_{i*})}{\partial q_{ik}} = \frac{\partial {}^w H_{ih}}{\partial q_{ik}} {}^{ih} r_{i*} + {}^w H_{ih} \underbrace{\frac{\partial {}^{ih} r_{i*}}{\partial q_{ik}}}_{=0} \quad (20)$$

and

$${}^w \dot{r}_{i*} = \frac{d}{dt} ({}^w H_{ih} {}^{ih} r_{i*}) = {}^w \dot{H}_{ih} {}^{ih} r_{i*} + {}^w H_{ih} \underbrace{\dot{{}^{ih} r_{i*}}}_{=0} \quad (21)$$

and hence only depend on the corresponding derivative of the actuator-dependent homogeneous transformation matrix.

Similarly, rotation rates also depend on homogeneous transformation matrices and its derivatives. Using the linear operator $\Gamma\{\star\}$ applied to $\star \in \mathbb{R}^{3 \times 3}$

$$\Gamma\{\star\} = \text{diag} \left(\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \star \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \right) \quad (22)$$

with the properties

$$\frac{\partial \Gamma\{\star\}}{\partial q} = \Gamma\left\{\frac{\partial \star}{\partial q}\right\} \quad (23a)$$

$$\frac{d \Gamma\{\star\}}{dt} = \Gamma\left\{\frac{d \star}{dt}\right\} \quad (23b)$$

and the corresponding homogeneous transformations (5), the rotation rate ${}^w \omega$ of the mass i can be derived [32] by

$${}^w \omega_i = \Gamma\{{}^w \dot{R}_{ih} {}^w R_{ih}^T\}. \quad (24)$$

B. Generalized Forces

For the purpose of better interpretation, bellows forces are divided in actuation forces $F_{Bi k,a}$ (index a) and passive forces $F_{Bi k,p}$ (index p)

$$F_{Bi k,a} = F_{Bi k,act}(p_{ik}) = A_p \cdot (p_{ik} - p_{amb}) \quad (25)$$

$$F_{Bi k,p} = F_{Bi k,spr}(q_{ik}) + F_{Bi k,dmp}(\dot{q}_{ik}). \quad (26)$$

The actuation forces result from the pressure difference inside the bellows and the ambient pressure working against the terminal surfaces A_p . The passive forces are the sum of the length-dependent longitudinal spring force and the velocity-dependent damping force.

With the bellows being always orthogonal to the head and base frame, the forces $F_{Bi k,*}$ transmitted by the bellows apply orthogonally to head and base at $r_{Bi k}$. Therefore, the forces in the local head and base coordinate frame have only entries in the (z)-direction. By serially stacking multiple sections on top of each other, the actuation and passive net forces $F_{e,Bi k,a}$ and $F_{e,Bi k,p}$

$${}^{ih} \mathbf{F}_{e,Bi k,*} = \begin{cases} [0 & 0 & F_{Bi k,*} - F_{B\{i+1\}k,*} & 0]^T & i < n \\ [0 & 0 & F_{Bi k,*} & 0]^T & i = n \end{cases} \quad (27)$$

at a sections head i result from the forces of the two adjacent sections if the section is not the final section n at the manipulator's head.

External forces in general can apply at every point, but in order to keep the equations as simple as possible, only external forces F_{Si} applying at the centers of mass r_{Si} are considered in the following, knowing that extending the resulting equations is easy to perform.

The generalized forces Q_j (17) depend on active forces. Active forces wF_e (index e) are both the external forces ${}^wF_{e,Si}$

$${}^wF_{e,Si} = {}^wH_{ih} {}^{ih}F_{Si,ext} \quad (28)$$

that apply at the center of mass and are zero if no external contact forces apply and the active forces of the bellows ${}^wF_{e,Bik}$

$${}^wF_{e,Bik} = {}^wF_{e,Bik,a} + {}^wF_{e,Bik,p} \quad (29)$$

consisting of actuation forces and passive forces with the actuation forces resulting from the pressure forces of the bellows (25), (27) and the passive forces from the dynamic spring and damper forces

$${}^wF_{e,Bik,a} = {}^wH_{ih} {}^{ih}F_{e,Bik,act} \quad (30)$$

$${}^wF_{e,Bik,p} = -{}^wH_{ih} ({}^{ih}F_{e,Bik,spr} + {}^{ih}F_{e,Bik,dmp}), \quad (31)$$

recalling that the active bellows forces ${}^wF_{e,Bik,*}$ are the resulting forces of a section pair (27) if the section is not the final section.

A generalized force (17)

$$Q_{ik} = \sum_{\xi=1}^n \left({}^wF_{e,S\xi}^T \frac{\partial {}^wH_{\xi h}}{\partial q_{ik}} {}^{\xi h}r_{S\xi} + \sum_{\chi=1}^3 {}^wF_{e,B\xi\chi}^T \frac{\partial {}^wH_{\xi h}}{\partial q_{ik}} {}^{B\xi\chi}r_{B\xi\chi} \right) \quad (32)$$

for its corresponding actuator state q_{ik} can be written as a sum of all projected (20) generalized forces including bellows forces (29) and forces applying at the center of mass (28). Note that gravitation and rotational spring forces due to bending are not included in the generalized forces but in the potential U .

C. Potential

The potential U_i

$$U_i = -m_i {}^w\mathbf{g}^T {}^w\mathbf{r}_{Si} + M_{i,bend}(\psi_i, \phi_i) \quad (33)$$

of a section i includes the gravitational potential of its mass m_i located at ${}^w\mathbf{r}_{Si}$ and rotational spring momentum $M_{i,bend}$ depending on the section's bending angle ψ_i (3) and the bending orientation ϕ_i . This momentum is introduced, as the longitudinal spring force of a bellows (26) derived from the single bellows model only covers the stiffness of its backbone against elongation, but not its width-depending stiffness against lateral applied forces. The resulting net momentum of three parallel bellows (see Fig. 7) disturbing each other's longitudinal stiffness is taken into account by the rotational spring momentum $M_{i,bend}$. This momentum also stabilizes a section to stand straight with zero curvature if no external forces apply.

The partial derivatives of the section's potential are

$$\frac{\partial U_i}{\partial q_{ik}} = -m_i {}^w\mathbf{g}^T \frac{\partial {}^w\mathbf{r}_{Si}}{\partial q_{ik}} + \frac{\partial M_{i,bend}}{\partial q_{ik}}. \quad (34)$$

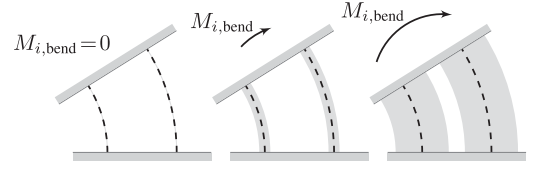


Fig. 7. Introduced bending stiffness $M_{i,bend}$ of section i considers the lateral stiffness due to the nonzero width of the bellows that forces single bellows to stand straight if no external forces are applied.

The sum of all n sections' potentials derived by the generalized coordinate q_{ik}

$$\frac{\partial U}{\partial q_{ik}} = \sum_{\xi=1}^n \left(-m_{\xi} {}^w\mathbf{g}^T \frac{\partial {}^wH_{\xi h}}{\partial q_{ik}} {}^{\xi h}r_{S\xi} + \frac{\partial M_{\xi,bend}}{\partial q_{ik}} \right) \quad (35)$$

describes the influence of the global potential on the generalized coordinate that is used for Euler-Lagrange equation (15).

D. Kinetic Energy

Using the homogeneous derivative (21) to describe the velocities of n point masses, the kinetic energy (16) with respect to the global reference system $\mathcal{R}_w$ is

$$T = T_{trans} + T_{rot} \quad (36)$$

with

$$T_{trans} = \frac{1}{2} \sum_{\xi=1}^n m_{\xi} \left({}^w\dot{\mathbf{H}}_{\xi h} {}^{\xi h}r_{S\xi} \right)^T \left({}^w\dot{\mathbf{H}}_{\xi h} {}^{\xi h}r_{S\xi} \right) \quad (37)$$

$$T_{rot} = \frac{1}{2} \sum_{\xi=1}^n \Gamma \left\{ {}^w\dot{\mathbf{R}}_{\xi h} {}^w\mathbf{R}_{\xi h}^T \right\}^T {}^wJ_{S\xi} \Gamma \left\{ {}^w\dot{\mathbf{R}}_{\xi h} {}^w\mathbf{R}_{\xi h}^T \right\} \quad (38)$$

and the moment of inertia expressed in world coordinates

$${}^wJ_{S\xi} = {}^w\mathbf{R}_{\xi h} {}^{\xi h}J_{S\xi} {}^w\mathbf{R}_{\xi h}^T. \quad (39)$$

To apply the Euler-Lagrange equations (15), the partial derivatives of (36) with respect to all generalized coordinates q_{ik} are needed. Using the chain rule and the identity⁴

$$\frac{\partial \mathbf{H}}{\partial q_j} = \frac{\partial \dot{\mathbf{H}}}{\partial \dot{q}_j}, \quad (40)$$

the derivatives of the translational kinetic energy (37) for constant masses m_i are

$$\frac{\partial T_{trans}}{\partial q_{ik}} = \sum_{\xi=1}^n m_{\xi} \left({}^w\dot{\mathbf{H}}_{\xi h} {}^{\xi h}r_{S\xi} \right)^T \left(\frac{\partial {}^w\dot{\mathbf{H}}_{\xi h}}{\partial q_{ik}} {}^{\xi h}r_{S\xi} \right) \quad (41)$$

$$\frac{\partial T_{trans}}{\partial \dot{q}_{ik}} = \sum_{\xi=1}^n m_{\xi} \left({}^w\dot{\mathbf{H}}_{\xi h} {}^{\xi h}r_{S\xi} \right)^T \left(\frac{\partial {}^w\mathbf{H}_{\xi h}}{\partial q_{ik}} {}^{\xi h}r_{S\xi} \right) \quad (42)$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial T_{trans}}{\partial \dot{q}_{ik}} &= \sum_{\xi=1}^n m_{\xi} \left({}^w\ddot{\mathbf{H}}_{\xi h} {}^{\xi h}r_{S\xi} \right)^T \left(\frac{\partial {}^w\mathbf{H}_{\xi h}}{\partial q_{ik}} {}^{\xi h}r_{S\xi} \right) \\ &+ \frac{\partial T_{trans}}{\partial q_{ik}}. \end{aligned} \quad (43)$$

⁴A short proof can be found in Appendix A.

The derivatives of the homogeneous transformations $\mathbf{H}$ used in (41)–(43) are presented in Appendix B. Similarly, the derivatives of the rotation matrix $\mathbf{R}$ can be implemented, as the identity of

$$\frac{\partial \mathbf{R}}{\partial q} = \frac{\partial \dot{\mathbf{R}}}{\partial \dot{q}} \quad (44)$$

follows (40). To compute the derivatives of the rotational kinetic energy (38) similar to (41)–(43), the properties of the linear operator Γ (23) are used, such that the derivatives of T_{rot} for constant inertia ${}^{\xi h} \mathbf{J}_{S\xi}$ are

$$\begin{aligned} \frac{\partial T_{\text{rot}}}{\partial q_{ik}} &= \sum_{\xi=1}^n \Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\}^T \mathbf{R}^{\xi h} \mathbf{J}_{S\xi} \left(\frac{\partial \mathbf{R}^T}{\partial q_{ik}} \Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\} \right. \\ &\quad \left. + \mathbf{R}^T \Gamma \left\{ \frac{\partial \dot{\mathbf{R}}}{\partial q_{ik}} \mathbf{R}^T + \dot{\mathbf{R}} \frac{\partial \mathbf{R}^T}{\partial q_{ik}} \right\} \right) \end{aligned} \quad (45)$$

$$\frac{\partial T_{\text{rot}}}{\partial \dot{q}_{ik}} = \sum_{\xi=1}^n \Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\}^T \mathbf{R}^{\xi h} \mathbf{J}_{S\xi} \mathbf{R}^T \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_{ik}} \mathbf{R}^T \right\} \quad (46)$$

$$\begin{aligned} \frac{d}{dt} \frac{\partial T_{\text{rot}}}{\partial \dot{q}_{ik}} &= \sum_{\xi=1}^n \left(\Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\}^T \dot{\mathbf{R}} + \Gamma \left\{ \ddot{\mathbf{R}} \mathbf{R}^T + \dot{\mathbf{R}} \dot{\mathbf{R}}^T \right\}^T \mathbf{R} \right) \\ &\quad {}^{\xi h} \mathbf{J}_{S\xi} \mathbf{R}^T \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_{ik}} \mathbf{R}^T \right\} \\ &\quad + \Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\}^T \mathbf{R}^{\xi h} \mathbf{J}_{S\xi} \\ &\quad \left(\dot{\mathbf{R}}^T \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_{ik}} \mathbf{R}^T \right\} - \frac{\partial \mathbf{R}^T}{\partial q_{ik}} \Gamma \left\{ \dot{\mathbf{R}} \mathbf{R}^T \right\} \right) \\ &\quad + \frac{\partial T_{\text{rot}}}{\partial q_{ik}} \end{aligned} \quad (47)$$

using the abbreviation $\mathbf{R}$ for ${}^w \mathbf{R}_{\xi h}$ and its derivatives.

E. Resulting Dynamic Model

By inserting the generalized forces (32) for all generalized coordinates q_j and the derivatives of potential (35) and the kinetic energy (41), (43), (45), (47) into the Euler–Lagrange equation (15), $3n$ coupled second-order differential equations are obtained. By resorting, the differential equations can be written in matrix form as

$$\mathbf{M}(q) \ddot{\mathbf{q}} + \mathbf{C}(q, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{N}(q, \dot{\mathbf{q}}) = \boldsymbol{\tau}(\mathbf{p}), \quad (48)$$

in which the active, controllable forces ${}^w \mathbf{F}_{Bik,a}$ are part of the inhomogeneous input force vector $\boldsymbol{\tau}(\mathbf{p})$ of the differential equation. The resulting mass matrix $\mathbf{M} \in \mathbb{R}^{3n \times 3n}$ and the matrix of Coriolis and centrifugal gains $\mathbf{C} \in \mathbb{R}^{3n \times 3n}$ are defined by the entry-wise formulas

$$\begin{aligned} \mathbf{M}_{(r,c)} &= \sum_{\xi=s}^n \left(m_{\xi} \left(\frac{\partial \mathbf{H}}{\partial q_c} {}^{\xi h} \mathbf{r}_{S\xi} \right)^T \frac{\partial \mathbf{H}}{\partial q_r} {}^{\xi h} \mathbf{r}_{S\xi} \right. \\ &\quad \left. + \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_c} \mathbf{R}^T \right\}^T {}^w \mathbf{J}_{S\xi} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_r} \mathbf{R}^T \right\} \right) \end{aligned} \quad (49)$$

$$\begin{aligned} \mathbf{C}_{(r,c)} &= \sum_{\xi=s}^n \left(m_{\xi} \left(\frac{\partial \dot{\mathbf{H}}}{\partial q_c} {}^{\xi h} \mathbf{r}_{S\xi} \right)^T \frac{\partial \mathbf{H}}{\partial q_r} {}^{\xi h} \mathbf{r}_{S\xi} \right. \\ &\quad + \Gamma \left\{ \frac{\partial \dot{\mathbf{R}}}{\partial q_c} \mathbf{R}^T \right\}^T {}^w \mathbf{J}_{S\xi} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_r} \mathbf{R}^T \right\} \\ &\quad + \sum_{\gamma=1}^{3\xi} \dot{q}_{\gamma} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_c} \frac{\partial \mathbf{R}^T}{\partial q_{\gamma}} \right\}^T {}^w \mathbf{J}_{S\xi} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_r} \mathbf{R}^T \right\} \\ &\quad + \sum_{\gamma=1}^{3\xi} \dot{q}_{\gamma} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_c} \mathbf{R}^T \right\}^T \frac{\partial \mathbf{R}}{\partial q_{\gamma}} {}^{\xi h} \mathbf{J}_{S\xi} \mathbf{R}^T \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_r} \mathbf{R}^T \right\} \\ &\quad + \sum_{\gamma=1}^{3\xi} \dot{q}_{\gamma} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_c} \mathbf{R}^T \right\}^T \mathbf{R}^{\xi h} \mathbf{J}_{S\xi} \frac{\partial \mathbf{R}^T}{\partial q_{\gamma}} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_r} \mathbf{R}^T \right\} \\ &\quad \left. - \sum_{\gamma=1}^{3\xi} \dot{q}_{\gamma} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_c} \mathbf{R}^T \right\}^T \mathbf{R}^{\xi h} \mathbf{J}_{S\xi} \frac{\partial \mathbf{R}^T}{\partial q_r} \Gamma \left\{ \frac{\partial \mathbf{R}}{\partial q_{\gamma}} \mathbf{R}^T \right\} \right) \end{aligned} \quad (50)$$

with $r, c \in [1, 3n]$ indicating the row and column index of the matrices and using the abbreviation $\mathbf{H}$ for ${}^w \mathbf{H}_{\xi h}$ and its derivatives ($\mathbf{R}$ for ${}^w \mathbf{R}_{\xi h}$ respectively). The start index of the sums is defined by

$$s = \text{section of}(\max(r, c)). \quad (51)$$

The last three rows of (50) result from nonconstant ${}^w \mathbf{J}_{S\xi}$ due to the rotation of constant ${}^{\xi h} \mathbf{J}_{S\xi}$ in the world reference frame (39). The homogeneous force vector $\mathbf{N} \in \mathbb{R}^{3n}$

$$\begin{aligned} \mathbf{N}_{(r)} &= \sum_{\xi=1}^n \left(\frac{\partial M_{r,\text{bend}}}{\partial q_r} - \left({}^w \mathbf{F}_{e,S}^T + m_{\xi} {}^w \mathbf{g}^T \right) \frac{\partial {}^w \mathbf{H}_{\xi h}}{\partial q_r} {}^{\xi h} \mathbf{r}_{S\xi} \right) \\ &\quad - \sum_{\xi=1}^n \sum_{\chi=1}^3 {}^w \mathbf{F}_{e,B\xi\chi,a}^T \frac{\partial {}^w \mathbf{H}_{\xi h}}{\partial q_r} {}^{\xi h} \mathbf{r}_{B\xi\chi} \end{aligned} \quad (52)$$

contains stiffness and damping properties, as well as external and gravitational influences, while the pressure-dependent inhomogeneous force vector $\boldsymbol{\tau}(\mathbf{p}) \in \mathbb{R}^{3n}$ is defined as

$$\boldsymbol{\tau}_{(r)} = \sum_{\xi=1}^n \sum_{\chi=1}^3 {}^w \mathbf{F}_{e,B\xi\chi,a}^T \frac{\partial {}^w \mathbf{H}_{\xi h}}{\partial q_r} {}^{\xi h} \mathbf{r}_{B\xi\chi} \quad (53a)$$

$$= \sum_{\xi=1}^n \sum_{\chi=1}^3 {}^{\xi h} \mathbf{F}_{e,B\xi\chi,a}^T {}^w \mathbf{H}_{\xi h}^T \frac{\partial {}^w \mathbf{H}_{\xi h}}{\partial q_r} {}^{\xi h} \mathbf{r}_{B\xi\chi} \quad (53b)$$

with forces in the global (53a) or local (53b) reference frame (19). As the pressures of the bellows are the inputs of the mechanical model but their progressions cannot be chosen arbitrarily, dynamic models for the pneumatic behavior of the individual bellows are necessary as well. The pneumatic model of a bellows [30] based on models of pneumatic cylinders [33]–[35] is taken from existing models.

This analytical model can be efficiently programmed and implemented by computing reoccurring terms only once, e.g., $\frac{\partial \mathbf{H}}{\partial q_{\star}}$ in (49) and (50). This reduces the complexity of the analytical model and allows a practicable handling of the derivatives.

If the model is extended by adding additional external forces, these additional forces need to be separated into passive forces assigned to N , and actuated forces assigned to τ , although the latter should be rather uncommon.

IV. MODEL IDENTIFICATION

With the structural model derived (48)–(53), it can be programmed efficiently, but in order to run simulations, parameters need to be identified. All kinematic parameters like the distance of the bellows' center to the section's backbone r_{Bi} (7b), (18b) or the sizes of the terminal surfaces A_p of the bellows (25) can be taken from the CAD model or have to be identified beforehand in order to obtain the kinematic model. The position ${}^{ih}r_{Si}$ of each section's mass point in its local head frame (18a) is placed at the center

$${}^{ih}r_{Si} = [0, 0, 0, 1]^T \quad (54)$$

of the head's profile. The remaining dynamic parameters that have to be identified are

- the longitudinal stiffness or spring force $F_{Bik,spr}(q_{ik})$ of the bellows depending on its elongation,
- the longitudinal damping force $F_{Bik,dmp}(\dot{q}_{ik})$ of the bellows depending on its velocity,
- each section's bending momentum $M_{i,bend}(\psi_i)$ depending on the bending angle of the section,
- the nominal mass m_i of each section, and
- the corresponding constant moments of inertia ${}^{ih}J_{Si}$ of each mass.

In the following, these functions and constant parameters are identified on the Bionic Handling Assistant with appropriate experiments and measurements.

A. Identification of the Nominal Section Mass

The point mass m_i of a section is an approximation of the real mass distributed over the whole section. A sufficient approximation of the section's mass can be obtained by weighing the individual sections before assembling. Other masses being moved like tubes and wire cables are below 1 % of the mass of a section's polyamide structure and therefore neglected.

B. Identification of Inertia

The inertia of each section can be obtained by a static model from CAD or approximated by a solid cylinder of average section height $\ell_{i,avg}$ and radius $r_{Bi} + r_{bi}$ as sum of bellow's backbone distance plus bellow's radius. The local inertia tensor results in

$${}^{ih}J_{Si} = \begin{bmatrix} {}^{ih}J_{Si,x} & 0 & 0 \\ 0 & {}^{ih}J_{Si,y} & 0 \\ 0 & 0 & {}^{ih}J_{Si,z} \end{bmatrix} \quad (55)$$

with the moments of inertia defined as

$${}^{ih}J_{Si,x} = \frac{1}{4} m_i (r_{Bi} + r_{bi})^2 + \frac{1}{12} m_i \ell_{i,avg}^2, \quad (56a)$$

$${}^{ih}J_{Si,y} = \frac{1}{4} m_i (r_{Bi} + r_{bi})^2 + \frac{1}{12} m_i \ell_{i,avg}^2, \quad (56b)$$

$${}^{ih}J_{Si,z} = \frac{1}{2} m_i (r_{Bi} + r_{bi})^2. \quad (56c)$$

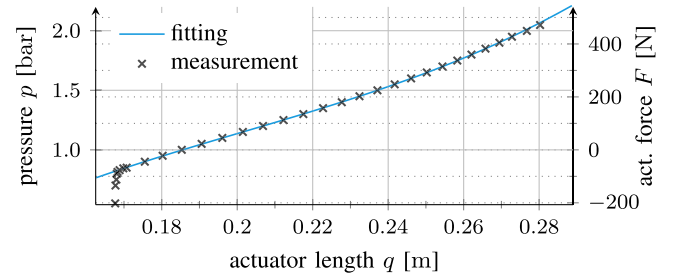


Fig. 8. Identification of the nonlinear stiffness behavior by measuring steady-state pressures for several elongations.

However, an approximation of the maximum moment of inertia can be obtained by assuming a hollow cylinder with maximal section length $\ell_{i,max}$ and radius $r_{Bi} + r_{bi}$

$${}^{ih}J_{Si,x,max} = \frac{1}{2} m_i (r_{Bi} + r_{bi})^2 + \frac{1}{12} m_i \ell_{i,max}^2 \quad (57a)$$

$${}^{ih}J_{Si,y,max} = \frac{1}{2} m_i (r_{Bi} + r_{bi})^2 + \frac{1}{12} m_i \ell_{i,max}^2 \quad (57b)$$

$${}^{ih}J_{Si,z,max} = m_i (r_{Bi} + r_{bi})^2 \quad (57c)$$

that is used for the calculation of a maximum bound of rotational energies.

These constant moments of inertia in the local frame $\mathcal{R}_{ih}$ can be transformed into $\mathcal{R}_w$ using (39) and the rotation matrix ${}^wR_{ih}$ of the corresponding homogeneous transformation (5). The implementation of variable local inertia matrices ${}^{ih}J_{Si}$ depending on the configuration states (section length, curvature, and orientation) is not worth the effort due to its small influence at possible motions.

C. Identification of Longitudinal Stiffness

The longitudinal stiffness of the bellows is represented by the bellows' spring force $F_{Bik,spr}(q_{ik})$, which is a nonlinear function depending on the length of each actuator (26). As a steady-state force, it can be identified by elongating all bellows of each section evenly such that the sections curvature stays zero (7). The spring forces are obtained by calculating actuation forces from measured steady-state pressures (25) and subtracting gravitational influences (see Fig. 8).

D. Identification of Bending Stiffness

The bending stiffness of a section represented by the section's bending momentum $M_{i,bend}$ depends on the bending angle ψ_i (3) and influences the potential of a section (33). Similar to the identification process of the longitudinal stiffness, the bending stiffness can be obtained via the measurement of steady-state pressures. For this purpose, actuators 2 and 3 of each section are evenly actuated against actuator 1 such that the orientation ϕ is either 0 or π (7a). Measured steady-state pressures are adjusted by subtracting longitudinal stiffness pressures and gravitational effects. The obtained relative bending pressures can be approximated by a linear function of the bending angle (see Fig. 9) although a variance of relative pressures exists caused by the small creeping of the section's material. With geometric parameters, the linear momentum function can be computed.

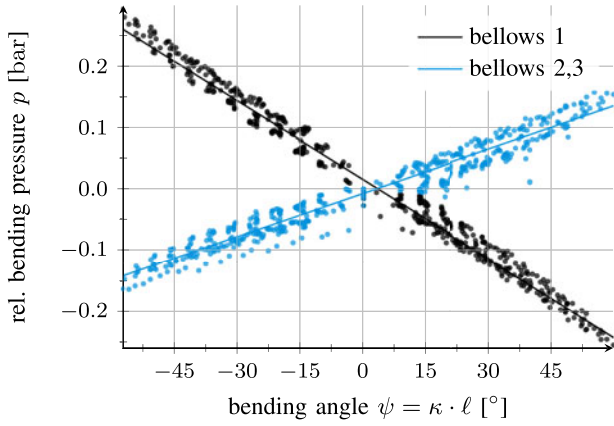


Fig. 9. Linear fitting of the relative bending pressures versus bending angle of bellows 1 working against bellows 2 and 3. Negative curvature indicates bending in the other direction $\phi = \pi$.

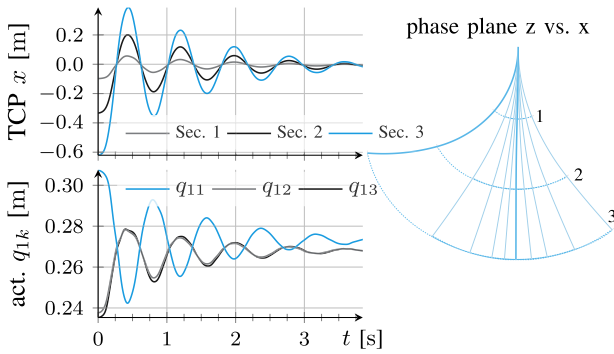


Fig. 10. Lateral swing triggered by contact loss of an external force to identify the manipulator's damping properties. Oscillations are visible on lateral deflection of the three sections x and a selection of actuator lengths q .

E. Identification of Longitudinal Damping

Due to small velocities, the damping forces $F_{Bik,dmp}(\dot{q}_{ik})$ of each bellows (26) have a small impact on dynamics if the manipulator is driven only by actuator force. However, on swings and oscillations triggered by jumps in external forces, e.g., a loss of contact (see Fig. 10), damping forces characterize the frequency and the decay of the oscillation. Using a least-squares fit, linear damping parameters can be identified from manipulator oscillations (see Fig. 10).

V. MODEL REDUCTION

When running dynamic simulations, a detailed model is desired. However, small errors are acceptable if the model is used for model-based control such that small errors can be compensated by feedback control and the computation time can be reduced significantly—e.g., for inverse dynamics control and model-based decoupling of the actuators using feedback-linearization [36].

In the following, two model reduction approaches are presented: First, rotational kinetic energies are neglected for the computation of the dynamic matrices $\mathbf{M}$ and $\mathbf{C}$. Second, the influence of the Coriolis matrix $\mathbf{C}$ on the dynamic model (48) is investigated and how the model changes if Coriolis forces are neglected.

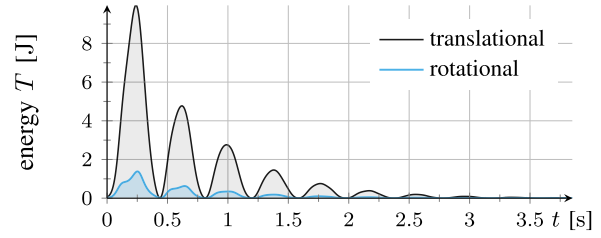


Fig. 11. Comparison of the measured translational energy and upper bounded rotational energy (57) of a lateral swing (see Fig. 10) with maximal rotation.

A. Neglecting Rotational Kinetic Energies

Rotational kinetic energies T_{rot} (16) are high on classic industrial robots with rigid arms and rotational joints as all motions are based on rotations only, especially due to the decreasing inertia of outer arms. Continuum manipulators do not have this kind of rotational joint. Rotations are comparably slow and can be imposed only by contrary extending and contracting bellows of a single section. Furthermore, the main motions of the imposed section mass and the attached sections will be of translational manner as, for example, a bending of the first section results in a small rotation of the upper sections but also in a large translation of the upper sections due to their leverage factor. Using a maximum approximation of each section's inertia (57), the rotational kinetic energy of a lateral swing (see Fig. 10) with maximal rotational velocities is significantly lower than the translational kinetic energy (see Fig. 11). Compared with this worst-case scenario, other motions generated by actuation force instead of external contact loss contain even less rotational kinetic energies.

Therefore, rotational kinetic energies can be neglected [29] for some purposes, especially for model-based control where computation time is important and small errors can be compensated by feedback control. This leads to a large simplification of the dynamic mass matrix $\mathbf{M}$ and the Coriolis matrix $\mathbf{C}$ such that the main dynamics are reduced to

$$\mathbf{M}_{(r,c)} = \sum_{\xi=s}^n m_{\xi} \left(\frac{\partial^w \mathbf{H}_{\xi h} \xi_h}{\partial q_c} \mathbf{r}_{S\xi} \right)^T \frac{\partial^w \mathbf{H}_{\xi h} \xi_h}{\partial q_r} \mathbf{r}_{S\xi} \quad (58)$$

$$\mathbf{C}_{(r,c)} = \sum_{\xi=s}^n m_{\xi} \left(\frac{\partial^w \dot{\mathbf{H}}_{\xi h} \xi_h}{\partial q_c} \mathbf{r}_{S\xi} \right)^T \frac{\partial^w \mathbf{H}_{\xi h} \xi_h}{\partial q_r} \mathbf{r}_{S\xi} \quad (59)$$

instead of (49) and (50). The homogeneous force vector $\mathbf{N}$ and the pressure-dependent inhomogeneous force vector $\boldsymbol{\tau}(\mathbf{p})$ do not depend on kinetic energies and therefore remain the same.

B. Neglecting Coriolis and Centrifugal Forces

The computation of the Coriolis and centrifugal gain matrix $\mathbf{C}$ in the dynamic equation (48) is computationally very heavy even if rotational kinetic energies are neglected. The high computational burden is caused by the dependence on the partial derivative $\partial_h^w \dot{\mathbf{H}}_{\xi} / \partial q_j$ that is used only in $\mathbf{C}$ (50), (59) but takes about half of the computation time. With small velocities $\dot{\mathbf{q}} \approx 0$, however, this term becomes very small ((71) in Appendix) and

TABLE II
SELECTED PARAMETERS OF THE EXAMPLE MANIPULATOR

Symbol		Value
Mass per section	[kg]	1.25
Pressure range (abs.)	[10 ⁵ Pa]	0.5 ... 2.0
Pressure force	[N]	-200 ... 500
Bellows length range	[m]	0.17 ... 0.29
Bellows volume	[10 ⁻³ m ³]	0.23 ... 0.57

the Coriolis matrix

$$C_{(r,c)} = \sum_{\xi=s}^n m_{\xi} \left(\frac{\partial^w \mathbf{H}_{\xi h} \xi^h \mathbf{r}_{S\xi}}{\partial q_c} \right)^T \frac{\partial^w \mathbf{H}_{\xi h} \xi^h \mathbf{r}_{S\xi}}{\partial q_r} \approx 0 \quad (60)$$

loses influence. If computation time is to be reduced, the Coriolis matrix of the dynamic model can be neglected

$$M(q) \ddot{q} + N(q, \dot{q}) = \tau(p) \quad (61)$$

depending on the required accuracy of the model.

Comparisons between measurements and simulation for both model reduction approaches are shown in the experimental validation next.

VI. EXPERIMENTAL VALIDATION

In order to test and verify the presented model and its approximations, measurements on the Bionic Handling Assistant were compared with open loop simulations of the model using the parameters from Table II.

Measurements are recorded using a DSpace 1007 PPC with 1 ms sampling time. Reference measurements were obtained from a pressure controlled manipulator. Note that negative pressures up to approximately -0.3 bar can be applied to the bellows using vacuum, in order to achieve higher curvatures—i.e., if two bellows of a section are pressurized, the third can be artificially held short.

Simulations were implemented in MATLAB/Simulink including the analytical derivatives of the homogeneous transformations ((66), Appendix B). Singularities in the mathematical formulation of the kinematic model were excluded numerically to distinguish between curvature κ close to zero (11) instead of (9) using switching boundary of $\kappa = 1 \cdot 10^{-6}$.

First, experiments with an actuated manipulator were conducted. Therefore, some bellows were actuated using a pressure trajectory, while the remaining were held at constant pressure. However, due to the mechanical coupling of the actuators, even the nonactuated bellows change their length. By changing only the pressure of the first bellows in the first section and in a second step the pressures of all bellows in the first section, two tilting motions are created (see Fig. 12). The second and third sections are extended in order to create a larger inertia.

These measurements can be compared with a simulation of the full model (49), (50) and a reduced model neglecting both rotational kinetic energies and Coriolis forces (58), (61). During pressure-driven motions, both models are good approximations of the measured trajectories. However, static models cannot describe the motions that well (see Fig. 13). A

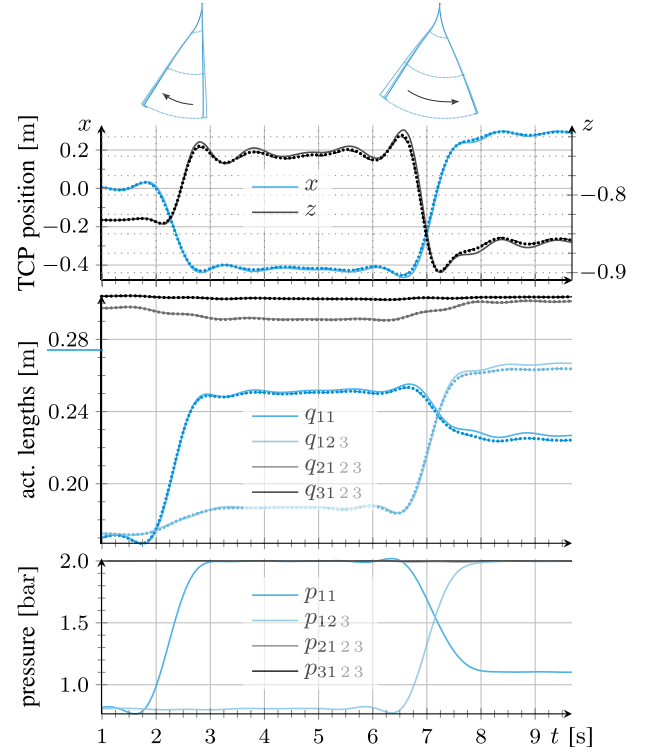


Fig. 12. Reference measurement data (—) versus simulated trajectories of a full model (·····) including rotational kinetic energies and Coriolis influences and a reduced model (---) without rotational kinetic energies and Coriolis influences.

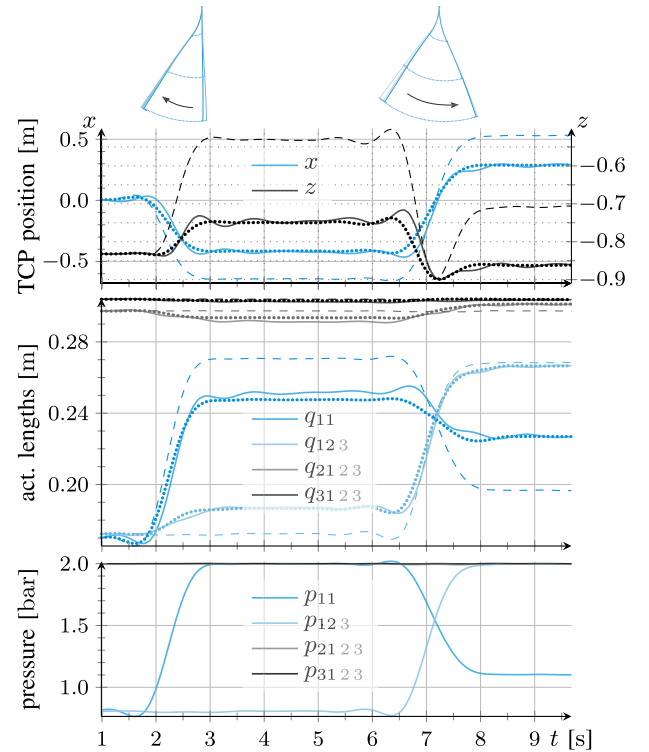


Fig. 13. Reference measurement data (—) versus simulated trajectories of a coupled static (·····) and decoupled static (---) model of a pressure driven trajectory in which section one is tilted.

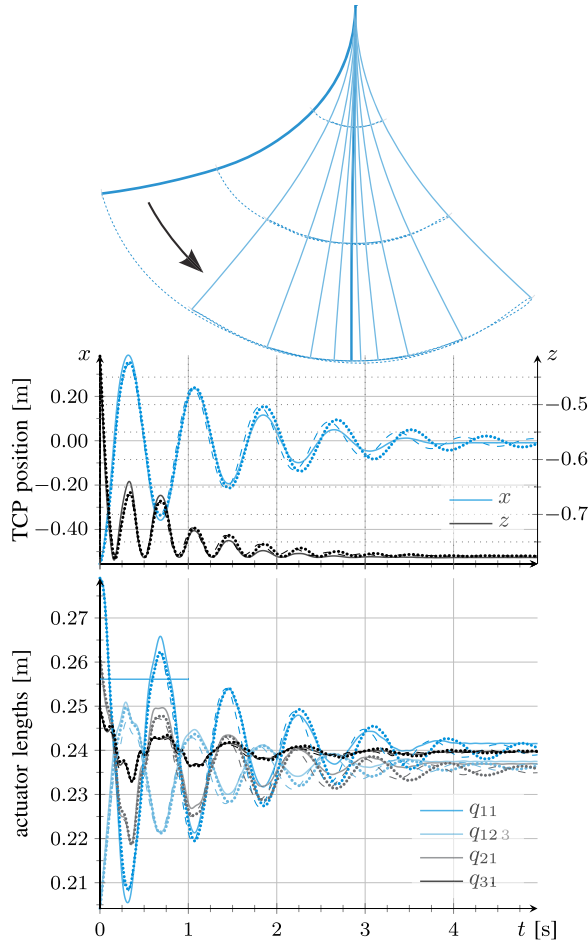


Fig. 14. Reference measurement data of a lateral swing (—) versus simulated trajectories of a full model (· · · · ·) including rotational kinetic energies and Coriolis influences and a reduced model (---) without rotational kinetic energies and Coriolis influences.

decoupled actuator model taking into account only translational spring forces against static pressures does not describe dynamic effects, gravity, and bending. A static model

$$N(q, \dot{q}) = \tau(p) \quad (62)$$

derived from (48) that takes into account also bending and gravity has also large errors during motions. Furthermore, the actuator lengths q cannot be computed explicitly from (62), and computing the solution implicitly takes even longer than the dynamic simulation.

Second, lateral swing motions similar to the damping identification (see Fig. 9) experiment are investigated. Actuated by the loss of an external contact force, the manipulator conducts a damped oscillation although the pressures in the bellows are not controlled (see Fig. 14). These motions cannot be simulated with static models, as the steady state does not change as long as the pressure in the bellows does not change. The developed dynamic models however—both full models and reduced models—can describe the motion quite well (see Fig. 14). The simulated frequency, amplitude, and damping of the tool center point (TCP) matches the reference measurement during the large oscillations at the beginning of the swing. However,

TABLE III
COMPARISON OF THE RELATIVE COMPUTATION TIME

Model w/	Rotations	Coriolis	Quality	Rel. Time
dynamic	x	x	++	1.000
dynamic	x	—	++	0.244
dynamic	—	x	+	0.798
dynamic	—	—	0	0.175
static (coupled) ^a			—	15.468
static (decoupled)			—	0.014

^a High computational effort due to an implicit nonlinear model.

simulation errors, especially in the actuator domain, exist due to limits of the lumped mass approximation of the manipulator. All further approximations made in the model reduction section—neglecting rotational kinetic energies and Coriolis forces—lead to very small changes in the simulation results, while the computation time decreases by about 80 %. A comparison of relative computation times with respect to the full dynamic model (48) is presented in Table III. It provides the computational effort of the methods independently of simulation step size, computational power, and simulation time.

VII. CONCLUSION

In this paper, a new approach for a dynamic model of bellows-actuated spatial continuum manipulators has been presented that includes couplings and interconnections between the parallel and serial arranged bellows. Doing this, each section is modeled as a single lumped mass point actuated by three parallel spring-damper systems. Using the constant curvature kinematic model, forces between the sections are transformed into a common world reference system such that the Euler-Lagrange equations for each actuator can be computed. With analytic derivatives of kinematics, the dynamic manipulator model can be easily implemented. After a parameter identification on the example manipulator, experiments show that the new model incorporates the interconnections between the bellows and that it describes the real manipulator behavior well. In order to reduce computation time, the complexity of the model can be reduced even further, if the desired accuracy is not as important—e.g., for inverse dynamics control and model-based decoupling of the actuators using feedback linearization.

APPENDIX A

PROVING THE IDENTITY OF HOMOGENEOUS DERIVATIVES

The identity of $\frac{\partial \mathbf{H}}{\partial q_j} = \frac{\partial \dot{\mathbf{H}}}{\partial \dot{q}_j}$ as stated in (40) is a common result from classical mechanics. However, due to its importance for the derivation of the dynamical model, a short proof will be presented here.

Proof: The identity of (40) can be validated by using the property

$$\frac{\partial r}{\partial q} = \frac{\partial \dot{r}}{\partial \dot{q}} \quad (63)$$

derived from the principle of virtual work [37] that is valid if r does not depend on $\dot{q}$. With the chain rule, the partial derivatives

of r (19) and $\dot{r}$ (21) are

$$\frac{\partial {}^w \mathbf{r}_{i*}}{\partial q_j} = \frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_j} {}^{ih} \mathbf{r}_{i*} + {}^w \mathbf{H}_{ih} \frac{\partial {}^{ih} \mathbf{r}_{i*}}{\partial q_j} \quad (64)$$

$$\frac{\partial {}^w \dot{\mathbf{r}}_{i*}}{\partial \dot{q}_j} = \frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial \dot{q}_j} {}^{ih} \mathbf{r}_{i*} + {}^w \dot{\mathbf{H}}_{ih} \frac{\partial {}^{ih} \mathbf{r}_{i*}}{\partial \dot{q}_j}. \quad (65)$$

With ${}^{ih} \mathbf{r}_{i*}$ being constant (18) and its partial derivatives being zero, the assumption (40) follows the property (63). ■

APPENDIX B

DERIVATIVES OF CONCATENATED HOMOGENEOUS TRANSFORMATIONS

The derivatives of the homogeneous transformation of a single section ${}^{ib} \mathbf{H}_{ih}(\mathbf{k}(\mathbf{q}))$ can be obtained by applying the chain rule, using the general mapping $\mathbf{f}_{\text{general}}$ (12) and the specific mapping $\mathbf{f}_{\text{specific}}$ (6) of the constant curvature approach. As the homogeneous transformation of a single section ${}^{ib} \mathbf{H}_{ih}$ only depends on the actuator states q_{ik} of this section, the derivative

$$\frac{\partial {}^{ib} \mathbf{H}_{ih}}{\partial q_{ik}} = \begin{cases} \mathbf{0} & i \neq \hat{i} \\ \frac{\partial {}^{ib} \mathbf{H}_{ih}}{\partial \mathbf{k}_i} \frac{\partial \mathbf{k}_i}{\partial q_{ik}} & i = \hat{i}, \end{cases} \quad (66)$$

is zero with respect to actuator states of other sections. Hence, the derivative of a global homogeneous transformation ${}^w \mathbf{H}_{ih}$

$$\frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{ik}} = \begin{cases} \mathbf{0} & i > \hat{i} \\ {}^w \mathbf{H}_{ib} \frac{\partial {}^{ib} \mathbf{H}_{ih}}{\partial q_{ik}} {}^{ih} \mathbf{H}_{ih} & i \leq \hat{i}, \end{cases} \quad (67)$$

is independent of sections i and actuators q_{ik} that are above section $\hat{i}$. Similar to (66), also the second partial derivatives of a section

$$\frac{\partial^2 {}^{ib} \mathbf{H}_{ih}}{\partial q_{\alpha\beta} \partial q_{ik}} = \begin{cases} \mathbf{0} & \alpha \neq \hat{i} \vee i \neq \hat{i} \\ \frac{\partial^2 {}^{ib} \mathbf{H}_{ih}}{\partial \mathbf{k}_i \partial \mathbf{k}_i} \frac{\partial^2 \mathbf{k}_i}{\partial q_{i\beta} \partial q_{ik}}, & \alpha = i = \hat{i} \end{cases} \quad (68)$$

with respect to actuator states of other sections are zero. Following (67) and defining without loss of generality $\alpha \leq i$, the second partial derivative

$$\frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{\alpha\beta} \partial q_{ik}} = \begin{cases} \mathbf{0} & i > \hat{i} \\ {}^w \mathbf{H}_{\alpha b} \frac{\partial {}^{\alpha b} \mathbf{H}_{\alpha h}}{\partial q_{\alpha\beta}} {}^{\alpha h} \mathbf{H}_{ib} \frac{\partial {}^{ib} \mathbf{H}_{ih}}{\partial q_{ik}} {}^{ih} \mathbf{H}_{ih} & \alpha < i \leq \hat{i} \\ {}^w \mathbf{H}_{ib} \frac{\partial^2 {}^{ib} \mathbf{H}_{ih}}{\partial q_{i\beta} \partial q_{ik}} {}^{ih} \mathbf{H}_{ih} & \alpha = i \leq \hat{i} \end{cases} \quad (69)$$

of a homogeneous transformation in the world reference frame can be derived.

The resulting time derivative of the time-invariant transformation ${}^w \mathbf{H}_{ih}$

$$\begin{aligned} {}^w \dot{\mathbf{H}}_{ih} &= \frac{d {}^w \mathbf{H}_{ih}}{dt} = \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi}} \dot{q}_{\xi\chi} + \underbrace{\frac{\partial {}^w \mathbf{H}_{ih}}{\partial t}}_{=0} \\ &= \sum_{\xi=1}^{\hat{i}} \sum_{\chi=1}^3 \frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi}} \dot{q}_{\xi\chi} \end{aligned} \quad (70)$$

only depends on partial derivatives and the corresponding actuator velocities, similar to the partial derivative of the time derivative

$$\begin{aligned} \frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial q_{ik}} &= \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi} \partial q_{ik}} \dot{q}_{\xi\chi} + \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi}} \underbrace{\frac{\partial \dot{q}_{\xi\chi}}{\partial q_{ik}}}_{=0} \\ &= \begin{cases} \mathbf{0}, & i > \hat{i} \\ \sum_{\xi=1}^{\hat{i}} \sum_{\chi=1}^3 \frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi} \partial q_{ik}} \dot{q}_{\xi\chi} & i \leq \hat{i} \end{cases} \end{aligned} \quad (71)$$

that equals the time derivative of the partial derivative

$$\begin{aligned} \frac{d}{dt} \frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{ik}} &= \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi} \partial q_{ik}} \dot{q}_{\xi\chi} + \underbrace{\frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{ik} \partial t}}_{=0} \\ &= \frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial q_{ik}}. \end{aligned} \quad (72)$$

The second time derivative of homogeneous transformations in the world reference frame ${}^w \ddot{\mathbf{H}}_{ih}$

$$\begin{aligned} {}^w \ddot{\mathbf{H}}_{ih} &= \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial \dot{q}_{\xi\chi}} \ddot{q}_{\xi\chi} + \sum_{\xi=1}^n \sum_{\chi=1}^3 \frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial q_{\xi\chi}} \dot{q}_{\xi\chi} + \underbrace{\frac{\partial {}^w \dot{\mathbf{H}}_{ih}}{\partial t}}_{=0} \\ &= \sum_{\xi=1}^{\hat{i}} \sum_{\chi=1}^3 \left(\frac{\partial {}^w \mathbf{H}_{ih}}{\partial q_{\xi\chi}} \ddot{q}_{\xi\chi} + \sum_{\alpha=1}^{\hat{i}} \sum_{\beta=1}^3 \frac{\partial^2 {}^w \mathbf{H}_{ih}}{\partial q_{\alpha\beta} \partial q_{\xi\chi}} \dot{q}_{\alpha\beta} \dot{q}_{\xi\chi} \right) \end{aligned} \quad (73)$$

used in (43) is obtained by using (40) and can be computed with partial derivatives of first (67) and second order (69) and the time derivatives of the actuator states $\mathbf{q}$.

Using the chain rule, the derivatives of the section's specific mapping (7) as well as the partial derivatives of the sections general mapping (9) used in (66) and (68) can be computed analytically [28] and then inserted into the formulas of the global derivatives. Similarly, all derivatives of rotation matrices ${}^w \mathbf{R}_{ih}$ can be computed, too.

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